

Homework 3 Solution

1. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation. Suppose that

$$T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}, \quad T \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}.$$

Find $T \begin{pmatrix} x \\ y \end{pmatrix}$ for every $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$.

Solution: We first note that the vectors $\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ are a basis of $\mathbb{R}^2$ as they are linearly independent. By the existence and uniqueness of a linear map determined by its values on a basis, we conclude that T exists. Let $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$.

Note that $\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \vec{v}_2 - \vec{v}_1$ and $\vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2\vec{v}_1 - \vec{v}_2$. It follows that

$$\begin{pmatrix} x \\ y \end{pmatrix} = x\vec{e}_1 + y\vec{e}_2 = x(\vec{v}_2 - \vec{v}_1) + y(2\vec{v}_1 - \vec{v}_2) = (2y - x) \cdot \vec{v}_1 + (x - y) \cdot \vec{v}_2$$

Consequently

$$\begin{aligned} T \begin{pmatrix} x \\ y \end{pmatrix} &= T((2y - x) \cdot \vec{v}_1 + (x - y) \cdot \vec{v}_2) \stackrel{\text{linear}}{=} (2y - x)T(\vec{v}_1) + (x - y)T(\vec{v}_2) \stackrel{\text{assumption}}{=} (2y - x) \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} + (x - y) \cdot \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 6y - 3x + 3x - 3y \\ x - 2y - x + y \\ 8y - 4x \end{pmatrix} = \begin{pmatrix} 3y \\ -y \\ 8y - 4x \end{pmatrix} \end{aligned}$$

2. Let $W = \{A \in M_n(\mathbb{R}) : A^t = A\}$ and let $H = \{A \in M_n(\mathbb{R}) : A^t = -A\}$.

Give an example of a linear operator $T : M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R})$ such that $\ker T = W$ and $\text{Im } T = H$.

Solution:

Consider the linear operator $T : M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R})$ defined by

$$T(A) = A - A^t, \quad \forall A \in M_n(\mathbb{R})$$

We first prove that T is a linear operator: let $\alpha, \beta \in \mathbb{R}$ and let $A, B \in M_n(\mathbb{R})$. Then

$$T(\alpha A + \beta B) = \alpha A + \beta B - (\alpha A + \beta B)^t = \alpha(A - A^t) + \beta(B - B^t) = \alpha T(A) + \beta T(B)$$

We now note that

$$\ker T = \{A \in M_n(\mathbb{R}) : T(A) = 0_n\} = \{A \in M_n(\mathbb{R}) : A - A^t = 0_n\} = \{A \in M_n(\mathbb{R}) : A = A^t\} = W.$$

Note that for every $A \in M_n(\mathbb{R})$ we have

$$(T(A))^t = (A - A^t)^t = A^t - (A^t)^t = A^t - A = -(A - A^t) = -T(A).$$

It follows that $\text{Im } T \subseteq H$. We now prove that $\text{Im } T = H$ by showing $\dim \text{Im } T = \dim H$:

We know that $M_n(\mathbb{R}) = W \oplus H$. By the first theorem of dimensions,

$$n^2 = \dim M_n(\mathbb{R}) = \dim W + \dim H - \dim(W \cap H) = \dim W - \dim H$$

By the second theorem of dimensions,

$$\dim \text{Im } T = \dim M_n(\mathbb{R}) - \dim \ker T = n^2 - \dim \ker T = n^2 - \dim W = \dim H.$$

We also know that $\dim W + \dim H = n^2$, thus, $\dim \text{Im } T = n^2 - \dim \ker T = n^2 - \dim W = \dim H$. Thus $\text{Im } T = H$.

3. Let V be a finite-dimensional vector space over a field F and let $T : V \rightarrow V$ be a linear operator. Suppose that $T \circ T = T$. Prove that

$$V = \ker T \oplus \text{Im } T$$

Solution:

We first prove $\ker T \cap \text{Im } T = \{0_V\}$.

As $\ker T \cap \text{Im } T$ is a subspace of V (intersection of subspaces) then $0_V \in \ker T \cap \text{Im } T = \{0_V\}$. Hence

$$\{0_V\} \subseteq \ker T \cap \text{Im } T.$$

We now prove the opposite inclusion. Let $w \in \ker T \cap \text{Im } T$. As $w \in \text{Im } T$ then there exists a $v \in V$ such that $w = T(v)$. It follows that

$$w = T(v) \stackrel{T \circ T = T}{=} T(T(v)) = T(w) \stackrel{w \in \ker T}{=} 0_V.$$

Thus $\ker T \cap \text{Im } T = \{0_V\}$. We now prove that $\ker T + \text{Im } T = V$. As any sum of subspaces is a subspace, then $\ker T + \text{Im } T \subseteq V$. To show equality, it is sufficient to prove equality of dimensions, namely

$$\dim(\ker T + \text{Im } T) = \dim V.$$

By the first theorem of dimensions,

$$\dim(\ker T + \text{Im } T) = \dim \ker T + \dim \text{Im } T - \dim(\ker T \cap \text{Im } T) = \dim \ker T + \dim \text{Im } T = \dim V,$$

where the last equality follows from the second theorem of dimensions.

4. Let V be a finite-dimensional vector space over the field F and let $T : V \rightarrow V$ be a linear operator. Let $v_1, v_2 \in V$. Suppose that $T(v_1), T(v_2)$ are linearly independent and that $T^2(v_1), T^2(v_2)$ are linearly dependent. Prove that $\dim V \geq 3$.

Solution:

As $T(v_1), T(v_2)$ are linearly independent and $T(v_1), T(v_2) \in \text{Im}(T)$ then $\dim \text{Im}(T) \geq 2$. As $T^2(v_1), T^2(v_2)$ are linearly dependent, then there exist $\alpha_1, \alpha_2 \in F$ such that $\alpha_1 T^2(v_1) + \alpha_2 T^2(v_2) = 0_V$ and such that $(\alpha_1, \alpha_2) \neq (0_F, 0_F)$. By linearity this implies that $\alpha_1 T(v_1) + \alpha_2 T(v_2) \in \ker T$. Note that $\alpha_1 T(v_1) + \alpha_2 T(v_2) \neq 0_V$ as $T(v_1), T(v_2)$ are linearly independent and $(\alpha_1, \alpha_2) \neq (0_F, 0_F)$. Thus, $\dim \ker T \geq 1$. By the second theorem of dimensions, $\dim(V) = \dim \ker(T) + \dim \text{Im}(T) \geq 3$.

5. Consider the linear operator $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y + z \\ x + y + z \end{pmatrix}, \quad \forall \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3.$$

Note: You do not have to prove T is a linear operator. Prove or disprove each of the following statements:

- (a) There exists a linear transformation $T : \mathbb{R}^9 \rightarrow \mathbb{R}^4$ for which $\dim \ker T \leq \dim \operatorname{Im} T$.

Solution:

False. ATC that there exists a linear transformation $T : \mathbb{R}^9 \rightarrow \mathbb{R}^4$ such that $\dim \ker T \leq \dim \operatorname{Im} T$. Since $\operatorname{Im} T$ is a subspace of $\mathbb{R}^4$ and $\dim \mathbb{R}^4 = 4$ it holds that $\dim \operatorname{Im} T \leq 4$. By the second theorem of dimensions we have

$$9 = \dim \mathbb{R}^9 = \dim \ker T + \dim \operatorname{Im} T \leq \dim \operatorname{Im} T + \dim \operatorname{Im} T \leq 4 + 4 = 8$$

Contradiction.

- (b) The vector spaces $W = \{A \in M_2(\mathbb{F}) : A^t = A\}$ and $\mathbb{F}_2[x]$ are isomorphic.

Solution: True. We know that $\dim W = \dim \mathbb{F}_2[x] = 3$. Thus, $W, \mathbb{F}_2[x]$ are two 3-dimensional vector spaces over the field $\mathbb{F}$, and therefore are isomorphic.

- (c) There exists a linear operator $S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $(S \circ T)(\vec{e}_3) = \vec{e}_2$ and $(S \circ T)(\vec{e}_2) = \vec{e}_3$.

Solution:

False. Assume towards a contradiction that such an operator exists. Then

$$\begin{aligned} \vec{e}_2 &= (S \circ T)(\vec{e}_3) = S \left(T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) = S \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \\ &= S \left(T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right) = (S \circ T)(\vec{e}_2) = \vec{e}_3. \end{aligned}$$

Thus $\vec{e}_2 = \vec{e}_3$. Contradiction.

- (d) There exists a linear operator $S : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that

$$(T \circ S)(\vec{e}_3) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \text{and} \quad (T \circ S)(\vec{e}_2) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Solution:

6. Let $A \in M_{m \times n}(\mathbb{R})$.

- (a) Let $\vec{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \in \mathbb{R}^m$. Suppose that $(\vec{y})^t \vec{y} = 0$. Prove that $\vec{y} = \vec{0}$.

Solution: We have

$$(\vec{y})^t \vec{y} = \sum_{i=1}^m y_i^2 = 0$$

and therefore $y_1 = \cdots = y_m = 0 \Rightarrow \vec{y} = \vec{0}$.

(b) Prove that $\text{Null}(A^t A) = \text{Null} A$.

Hint. Use item (a).

Solution: We first prove that $\text{Null} A \subseteq \text{Null}(A^t A)$. Let $\vec{x} \in \text{Null} A$. Then $A\vec{x} = \vec{0}$. Thus,

$$(A^t A) \cdot \vec{x} = A^t \cdot (A \cdot \vec{x}) = A^t \cdot \vec{0} = \vec{0}$$

and therefore $\vec{x} \in \text{Null}(A^t A)$.

We now prove the reverse direction, namely, that $\text{Null}(A^t A) \subseteq \text{Null} A$. Let $\vec{x} \in \text{Null}(A^t A)$. Then $(A^t A)\vec{x} = \vec{0}$. Denote $\vec{y} = A\vec{x}$. We want to prove that $\vec{y} = \vec{0}$. Multiplying both sides of the equality by $(\vec{x})^t$ gives

$$(\vec{y})^t \vec{y} = (A\vec{x})^t \cdot A\vec{x} = (\vec{x})^t A^t A\vec{x} = (\vec{x})^t \cdot \vec{0} = 0$$

and by item (a), $\vec{y} = \vec{0}$.

(c) Conclude that $\text{rank}(A^t A) = \text{rank} A$.

Hint. Use the rank-nullity theorem.

Solution: By the rank-nullity theorem, we have

$$\text{rank} A + \dim \text{Null} A = n$$

and

$$\text{rank}(A^t A) + \dim \text{Null}(A^t A) = n$$

It follows that

$$\text{rank}(A^t A) = n - \dim \text{Null}(A^t A) = n - \dim \text{Null} A = \text{rank} A$$